Three Admissible Directions and Three Spatial Dimensions: A Structural Bridge Candidate between $H_{\text{eff}} \simeq \mathbb{C}^3$ and the Horizontal Geometry of $\text{Heis}_3(\mathbb{R})$

Q7 of the Cosmochrony Spectral Geometry Programme

Jérôme Beau

Independent Researcher

jerome.beau@cosmochrony.org

2026

Abstract

Two independent sub-programmes of the Cosmochrony framework independently produce the integer three as a structural output. Paper Q5b [2] derives a four-dimensional effective Lorentzian geometry from the homogeneous dimension $D_{\text{hom}} = 4$ of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ (Bass–Guivarch), with three spatial directions: the two horizontal generators $\tilde{X}, \tilde{Y}$ and the central element $\tilde{Z} = [\tilde{X}, \tilde{Y}]$ (the last entering via sub-principal corrections, open problem Q5b-O2). Paper O23 [8] derives $\Sigma_c(n_3) = 3$ from quaternionic minimality; papers O28–O29 [1, 6] establish that the admissible projection space $H_{\text{eff}} = \mathbb{C}^3$ has rank $r_{\text{eff}} = 3$ and houses a spin-$\frac{1}{2}$ trajectory subspace V_ρ with $d_\rho = 2$. We identify a representation-theoretic mechanism that constrains any possible bridge between these two "3"s. First, we prove that the bridge cannot be a Lie algebra isomorphism: $\mathfrak{su}(2) \simeq \operatorname{Im}\mathbb{H}$ is semisimple, while $\mathfrak{heis}_3$ is nilpotent. Second, we show that $H_{\text{eff}} \simeq \text{Sym}^2(V_\rho)$ as an $\mathfrak{su}(2)$-module (the spin-1 representation), and that the irreducibility of $\text{Sym}^2(V_\rho)$ forces any equivariant map to the spatial sector to be unique up to positive scalar (Rigidity Lemma via Schur). Third, we prove a Schrödinger Quadratic Form Lemma: the $\mathfrak{su}(2)$ Casimir restricted to $\text{Sym}^2(V_\rho)$ pulls back, under the bridge map, to a positive-definite rank-3 quadratic form on the spatial block of the effective symbol, with the spin-weight grading $(\pm 1, 0)$ matching the Carnot grading $(1, 2)$ of $\mathfrak{heis}_3$ in a precise sense. These results stop short of identifying the two "3"s but they uniquely constrain the form any identification must take, and reduce the open question to a single computable criterion on $\sigma_2(L_{\text{eff}})$ that simultaneously resolves Q5b-O2 of [2]. A new analytical result completes the picture: we prove that the chirped discrete Fourier transform F_c (the metaplectic representative of the U(1) rotation) commutes with L_{Weil} for all (q, c), which implies that the cross-term part of the criterion holds structurally. Numerical verification on O25 checkpoints for $q \in \{61, 101, 151\}$ confirms the vanishing of cross terms for all tested pairs, and the residual isotropy gap $|A_H - A_z|$ is observed to decrease monotonically with q in the low-energy sector. All available analytical and numerical evidence is consistent with asymptotic isotropy $A_H = A_z$, reducing the identification problem to a single scalar convergence.

Keywords: Heisenberg group; Weil representation; spectral geometry; sub-Riemannian Laplacian; Carnot geometry; spin representations; $\mathfrak{su}(2)$ Casimir; metaplectic symmetry; Cosmochrony.

Contents

1 Introduction and Statement of the Problem

1.1 Context

The Cosmochrony programme derives physical observables from the spectral properties of admissible non-injective projections on the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. Two sub-programmes have now independently produced the integer 3 as a structural output.

The geometric "3" (Q-series). Paper Q5b [2] establishes that the BFS shell stratification of $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converges, in the pre-saturation regime, to the Carnot–Carathéodory foliation of $\mathrm{Heis}_3(\mathbb{R})$. The homogeneous dimension $D_{\mathrm{hom}} = 4$ (Bass–Guivarch) gives the limiting geometry the spectral dimension of a four-dimensional space. The corresponding effective metric has Lorentzian signature $(-,+,+,+)$, with one temporal direction τ and three spatial directions. Within the sub-Riemannian structure of $\mathrm{Heis}_3(\mathbb{R})$, these three spatial directions arise as: the two horizontal generators $\tilde{X}, \tilde{Y}$ (entering the sub-Riemannian Laplacian $\Delta_H = \tilde{X}^2 + \tilde{Y}^2$) and the central element $\tilde{Z} = [\tilde{X}, \tilde{Y}]$, which enters the effective metric via sub-principal corrections (open problem Q5b-O2 of [2]).

The admissible "3" (O-series). Paper O23 [8] proves that the Born–Infeld fibre admissibility constraint, combined with the Hurwitz-to-$\mathfrak{su}(2)$ bridge of SpectralCapacity [5], forces the admissible neutral sector to be the imaginary subspace $\operatorname{Im}\mathbb{H} \simeq \mathbb{R}^3$ of the minimal associative non-commutative algebra $\mathbb{H}$. This gives $\Sigma_c(n_3) = 3$ as a structural consequence, not an arithmetic input. Paper O28 [1] finds $H_{\mathrm{eff}} = \mathbb{C}^3$ (consistent with $\Sigma_c(n_3) = 3$) and rank $r_{\mathrm{eff}} = 3$ for the per-pair covariance. Paper O29 [6] identifies the trajectory subspace $V_\rho \subset H_{\mathrm{eff}}$ with $d_\rho = 2$ via the symmetric rank formula $r_{\mathrm{eff}} = d_\rho(d_\rho + 1)/2$, confirming the spin-$\frac{1}{2}$ sector. Paper O27 [4] proves unconditionally that any admissible morphism factors through $\mathfrak{su}(2)$, closing the implication chain emergence $\Rightarrow$ non-injectivity $\Rightarrow$ pair structure $\Rightarrow$ quadratic form $\Rightarrow$ $\mathfrak{su}(2)$.

1.2 The Problem

Both sub-programmes produce a three-dimensional effective structure. Their common substrate is the same group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, but they access it through different aspects: the Q-series through the BFS ball geometry, the O-series through the representation theory of the Weil blocks. The question studied in this paper is:

> *Are the three admissible directions of* $\operatorname{Im}\mathbb{H}$ *(O23) and the three spatial directions of the effective Lorentzian geometry (Q5b) the same three-dimensional structure, seen from two different vantage points?*

We show below that this question has a precise formulation, admits falsifiability criteria, but is *not yet settled* within the corpus. The present work does not prove the identification, but shows that if it holds, it is uniquely determined by irreducibility (Rigidity Lemma) and must satisfy a specific quadratic-symbol constraint (Criterion 5.1) whose verification would simultaneously resolve open problem Q5b-O2 of [2].

1.3 Conventions

Throughout, q is an odd prime and $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. We write $\pi_c : V_q \to H_{\mathrm{eff}} = \mathbb{C}^3$ for the admissible projection of the character-c Weil block, following the notation of O25–O29. The Heisenberg Lie algebra is $\mathfrak{heis}_3 = \mathrm{span}_{\mathbb{R}}\{X, Y, Z\}$ with $[X, Y] = Z$ and $[X, Z] = [Y, Z] = 0$. The Lie algebra $\mathfrak{su}(2)$ is spanned by $\{e_1, e_2, e_3\}$ with $[e_i, e_j] = \varepsilon_{ijk} e_k$ and is identified with $\operatorname{Im}\mathbb{H}$ via $e_1 \leftrightarrow \mathbf{i}$, $e_2 \leftrightarrow \mathbf{j}$, $e_3 \leftrightarrow \mathbf{k}$.

2 Two Independent Notions of Dimension Three

2.1 The Geometric Three: Carnot Grading and Spatial Directions

The Heisenberg group $\mathrm{Heis}_3(\mathbb{R})$ is a Carnot group of step 2. Its Lie algebra $\mathfrak{heis}_3$ carries a graded decomposition $\mathfrak{heis}_3 = V_1 \oplus V_2$ where $V_1 = \mathrm{span}\{X, Y\}$ is the horizontal distribution of Carnot degree 1 and $V_2 = \mathrm{span}\{Z\}$ is the central direction of Carnot degree 2. The sub-Riemannian Laplacian is $\Delta_H = X^2 + Y^2$ (acting on $\mathrm{Heis}_3(\mathbb{R})$ by left-invariant vector fields); it involves *only* the degree-1 generators. The homogeneous dimension is $D_{\mathrm{hom}} = 2 \cdot 1 + 1 \cdot 2 = 4$, giving BFS ball growth $|B_n| \sim Cn^4$ (Theorem 2.1 of [2]).

The effective Lorentzian metric of Theorem 6.1 of [2] is $g^{\mu\nu} = \mathrm{diag}(-A_\tau, A_H, A_H, A_z)$ in the basis $(\tau, \tilde{X}, \tilde{Y}, \tilde{Z})$, where:

- $A_\tau > 0$ governs the temporal sector,
- $A_H > 0$ governs the two horizontal directions (entering Δ_H at principal-symbol level),
- $A_z > 0$ governs the central direction $\tilde{Z}$, which enters the effective metric via sub-principal corrections to the symbol of L_{eff} (open problem Q5b-O2 of [2]).

Remark 2.1 (Horizontal vs. derived: two sub-strata of the spatial sector)**.** *The three spatial directions of $g^{\mu\nu}$ do* not *form a homogeneous set from the sub-Riemannian viewpoint. The directions $\tilde{X}, \tilde{Y}$ are horizontal (Carnot degree* 1*) and enter the effective operator at the level of the principal symbol. The direction $\tilde{Z} = [\tilde{X}, \tilde{Y}]$ is derived (Carnot degree* 2*) and is invisible to the principal symbol of Δ_H; it contributes to the effective metric only through sub-principal corrections. This two-level structure — degree-*1 *directions contributing at leading order, degree-*2 *direction contributing at sub-leading order — will play a key role in the bridge construction of Section 4, where it corresponds precisely to the distinction between spin-weight* ± 1 *and spin-weight* 0 *in the triplet* $\mathrm{Sym}^2(V_\rho)$*.*

The three-dimensional spatial sector of $g^{\mu\nu}$ is the direct sum $V_1 \oplus V_2 = \mathfrak{heis}_3/\mathbb{R}\tau$, where τ denotes the projective temporal direction. The "geometric 3" is $\dim_{\mathbb{R}}(V_1 \oplus V_2) = 3$.

2.2 The Admissible Three: Quaternionic Minimality

The admissible "3" arises at the representation-theoretic level. Paper O23 [8] proves that the unique minimal associative non-commutative algebra compatible with Born–Infeld fibre admissibility is $\mathbb{H}$, and that the admissible neutral directions correspond to $\mathrm{Im}\,\mathbb{H}$, which carries exactly three independent axes. The saturation threshold $\Sigma_c(n_3) = 3$ is a consequence of this quaternionic minimality, not an arithmetic input.

At the fibre level, the admissible projection $\pi_c : V_q \to H_{\mathrm{eff}}$ lands in $H_{\mathrm{eff}} = \mathbb{C}^3$, consistently with $\Sigma_c(n_3) = 3$ [1]. The per-pair covariance C_c in $\mathrm{End}(H_{\mathrm{eff}})$ has rank $r_{\mathrm{eff}} = 3$, filling H_{eff} completely [1]. Paper O29 [6] identifies the trajectory subspace $V_\rho \subset H_{\mathrm{eff}}$ with $d_\rho = 2$ via the symmetric rank formula; H_{eff} then appears as the space of degree-two symmetric tensors over V_ρ, i.e. $H_{\mathrm{eff}} \simeq \mathrm{Sym}^2_{\mathbb{C}}(V_\rho)$ as a vector space. The "admissible 3" is the complex dimension of H_{eff} (equivalently, the rank of C_c).

Remark 2.2 (Three distinct "3"s to be kept separate)**.** *The present paper distinguishes three a priori distinct occurrences of* 3 *in the programme: (i)* $\dim_{\mathbb{R}}(\mathfrak{heis}_3/\mathbb{R}\tau) = 3$ *(geometric, from Carnot structure), (ii)* $\dim_{\mathbb{R}}(\mathrm{Im}\,\mathbb{H}) = 3$ *(algebraic, from quaternionic minimality, O23), (iii)* $\dim_{\mathbb{C}}(H_{\mathrm{eff}}) = r_{\mathrm{eff}} = 3$ *(representation-theoretic, from O28). These coincide numerically but need not coincide structurally. Paper O23 [8] Section 1.2 already flags a similar cautionary remark within the O-series itself.*

3 The Algebraic Obstruction: $\mathfrak{su}(2) \not\simeq \mathfrak{heis}_3$

Proposition 3.1 (Non-isomorphism)**.** *The Lie algebras $\mathfrak{su}(2) \simeq \operatorname{Im}\mathbb{H}$ and $\mathfrak{heis}_3 = \mathfrak{heis}_3$ are non-isomorphic.*

Proof. The algebra $\mathfrak{su}(2)$ is semisimple (its Killing form is negative-definite); $\mathfrak{heis}_3$ is nilpotent of class 2 (its derived ideal $[\mathfrak{heis}_3, \mathfrak{heis}_3] = \operatorname{span}\{Z\}$ is central and non-trivial). A semisimple Lie algebra cannot be nilpotent. □

Proposition 3.1 shows that the admissible "3" (which lives in $\mathfrak{su}(2)$-space) and the geometric "3" (which lives in $\mathfrak{heis}_3$-space) cannot be identified by a Lie algebra isomorphism. Any bridge between them must therefore operate at a level where the two structures are *compatible without being isomorphic*: this is precisely the level of representations.

4 Structural Constraints on Any Bridge

4.1 The Symmetric Square Identification

Paper O29 [6] establishes $d_\rho = 2$, so $V_\rho \simeq \mathbb{C}^2$ as an $\mathfrak{su}(2)$-module: this is the unique irreducible spin-$\frac{1}{2}$ representation. The symmetric square $\operatorname{Sym}^2_{\mathbb{C}}(V_\rho)$ is the irreducible spin-1 representation of $\mathfrak{su}(2)$, of complex dimension 3. It has a canonical basis $\{e_+, e_0, e_-\}$ of weight eigenstates under the Cartan generator J_3: $J_3 e_\pm = \pm e_\pm$, $J_3 e_0 = 0$.

Proposition 4.1 (Symmetric square identification)**.** *As an $\mathfrak{su}(2)$-module, $H_{\mathrm{eff}} \simeq \operatorname{Sym}^2(V_\rho)$ is the unique irreducible three-dimensional representation of $\mathfrak{su}(2)$, with highest weight 1.*

Proof. Paper O28 [1] shows that the per-pair outer products $M_j = w_j w_j^T$ span $H_{\mathrm{eff}} = \mathbb{C}^3$, where $w_j = \pi_c(v_j) \in V_\rho \subset H_{\mathrm{eff}}$. The map $V_\rho \otimes V_\rho \to H_{\mathrm{eff}}$ given by $u \otimes v \mapsto uv^T$ has image contained in the complex symmetric matrices $\operatorname{Sym}^2_{\mathbb{C}}(V_\rho)$, since $M_j = w_j w_j^T$ is symmetric. Paper O29 [6] establishes $r_{\mathrm{eff}} = d_\rho(d_\rho+1)/2 = 3 = \dim \operatorname{Sym}^2(V_\rho)$, so the image spans $\operatorname{Sym}^2_{\mathbb{C}}(V_\rho)$. Since $V_\rho \simeq \mathbb{C}^2$ is the spin-$\frac{1}{2}$ representation, the symmetric square is the spin-1 representation by the standard Clebsch–Gordan decomposition $\mathbb{C}^2 \otimes \mathbb{C}^2 = \operatorname{Sym}^2 \oplus \wedge^2 = V_1 \oplus V_0$. □

Proposition 4.1 is not explicit in O28–O29, where $H_{\mathrm{eff}} = \mathbb{C}^3$ appears only as the ambient admissible projection space; the $\mathfrak{su}(2)$-module structure is inferred here from the symmetric rank formula.

4.2 Grading Correspondence

The spin-weight grading on $\operatorname{Sym}^2(V_\rho) = \operatorname{span}\{e_+, e_0, e_-\}$ decomposes the three basis states according to their J_3-eigenvalue $m \in \{+1, 0, -1\}$. The Carnot grading on $\mathfrak{heis}_3$ decomposes the generators according to their homogeneous degree $d \in \{1, 2\}$. These are distinct gradings on distinct spaces, but they are compatible in the following sense.

Remark 4.2 (Grading correspondence)**.** *The pair $(m, d) = (\pm 1, 1)$ (spin ± 1, Carnot degree 1) corresponds to the horizontal generators $\tilde{X}, \tilde{Y}$, and the pair $(m, d) = (0, 2)$ (spin 0, Carnot degree 2) corresponds to the central generator $\tilde{Z} = [\tilde{X}, \tilde{Y}]$. The pattern is: the two generators entering the principal symbol of Δ_H (at leading order) have $|m| = 1$, while the generator entering via sub-principal corrections (at sub-leading order) has $m = 0$. This grading correspondence is not a Lie algebra isomorphism (Proposition 3.1), but it shows that the two sub-strata of the spatial sector identified in Remark 2.1 map to the two spin-weight sectors of the spin-1 triplet in a grading-compatible way.*

4.3 Rigidity of Any Bridge: Schur's Lemma

The irreducibility of $\mathrm{Sym}^2(V_\rho)$ as an $\mathfrak{su}(2)$-module has an immediate and strong consequence for the possible forms of any bridge to the spatial sector.

Lemma 4.3 (Rigidity via Schur)**.** *Let W be a three-dimensional real vector space equipped with an $\mathfrak{su}(2)$-action that decomposes as $W \simeq \mathfrak{su}(2)$ (adjoint representation, i.e. spin-1 over $\mathbb{R}$). Any $\mathfrak{su}(2)$-equivariant $\mathbb{C}$-linear map $\varphi : \mathrm{Sym}^2(V_\rho) \to W \otimes_{\mathbb{R}} \mathbb{C}$ is either zero or a scalar multiple of the unique (up to phase) isomorphism. In particular, the image of φ is all of $W \otimes \mathbb{C}$, and the map is determined up to a single complex scalar.*

Proof. The module $\mathrm{Sym}^2(V_\rho)$ is irreducible of dimension 3 over $\mathbb{C}$. The complexification $W \otimes_{\mathbb{R}} \mathbb{C}$ of the real spin-1 representation is also irreducible of dimension 3 over $\mathbb{C}$ (the complexification of the adjoint of $\mathfrak{su}(2)$ is the spin-1 representation of $_2(\mathbb{C})$). By Schur's lemma, $\mathrm{Hom}_{\mathfrak{su}(2)}(\mathrm{Sym}^2(V_\rho), W \otimes \mathbb{C}) \simeq \mathbb{C}$: the space of equivariant maps is one-dimensional, so any non-zero equivariant map is an isomorphism, unique up to a non-zero scalar. □

Remark 4.4 (Consequence for the bridge)**.** *Lemma 4.3 implies that* there is at most one equivariant bridge *(up to overall normalisation) between $H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho)$ and the spatial sector of the effective metric. This is a strong rigidity result: the bridge, if it exists, is not a matter of choice. Its existence is the open question; its uniqueness is guaranteed by irreducibility.*

4.4 The Schrödinger Quadratic Form Lemma

We now compute the quadratic form induced by the $\mathfrak{su}(2)$-Casimir on $\mathrm{Sym}^2(V_\rho)$ and show that its pullback under any equivariant bridge is positive-definite on the spatial sector.

The Casimir operator of $\mathfrak{su}(2)$ on the spin-j irreducible representation is $\mathcal{C} = j(j+1) \cdot \mathrm{Id}$. On $\mathrm{Sym}^2(V_\rho)$ (spin $j = 1$), this gives $\mathcal{C}|_{\mathrm{Sym}^2} = 2 \cdot \mathrm{Id}$.

Lemma 4.5 (Schrödinger quadratic form)**.** *Let $\varphi : \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}$ be any $\mathfrak{su}(2)$-equivariant isomorphism to the spatial sector W_{sp} of the effective metric (the unique such map up to scalar, by Lemma 4.3). Then:*

1. *The quadratic form $Q_{\mathcal{C}}(v) := \langle v, \mathcal{C}v\rangle = 2\|v\|^2$ on $\mathrm{Sym}^2(V_\rho)$ is positive-definite of rank 3.*
2. *Under any equivariant bridge φ, the pullback $\varphi^* g_{\mathrm{sp}}$ (where $g_{\mathrm{sp}} = \mathrm{diag}(A_H, A_H, A_z)$ is the spatial block of $g^{\mu\nu}$) is also positive-definite of rank 3.*
3. *The induced map on quadratic forms is:*

$$\varphi^* g_{\mathrm{sp}} = \lambda_H \cdot Q_\pm + \lambda_z \cdot Q_0, \tag{1}$$

where $Q_\pm$ is the projection onto the spin-weight ± 1 eigenspace $\{e_+, e_-\}$, Q_0 is the projection onto the spin-weight 0 eigenspace $\{e_0\}$, and $\lambda_H = A_H/\|\varphi\|^2$, $\lambda_z = A_z/\|\varphi\|^2$ are positive constants.

Proof. Item (1) is immediate from $\mathcal{C} = 2 \cdot \mathrm{Id}$ and $\dim \mathrm{Sym}^2(V_\rho) = 3$. For items (2) and (3): the spatial metric $g_{\mathrm{sp}} = \mathrm{diag}(A_H, A_H, A_z)$ in the basis $(\tilde{X}, \tilde{Y}, \tilde{Z})$ preserves the sub-strata of Remark 2.1: it acts by A_H on $V_1 = \mathrm{span}\{\tilde{X}, \tilde{Y}\}$ (Carnot degree 1) and by A_z on $V_2 = \mathrm{span}\{\tilde{Z}\}$ (Carnot degree 2). The absence of cross terms between V_1 and V_2 in g_{sp} follows from the U(1) symmetry generated by J_3: the rotation $\tilde{X} \leftrightarrow \tilde{Y}$ (sending $e_+ \leftrightarrow e_-$) preserves g_{sp} (since $A_H = A_H$) and fixes e_0 (and hence $\tilde{Z}$), so no J_3-equivariant quadratic form can mix the weight-± 1 sector with the weight-0 sector. This U(1) argument holds independently of the existence of the full bridge φ; it is a consequence of the isotropic admissible regime assumed in Theorem 6.1 of [2]. Under the grading correspondence of Remark 4.2, V_1 maps to the spin-weight ± 1 sector $\{e_+, e_-\}$ and V_2 maps to the spin-weight 0 sector $\{e_0\}$. The pullback $\varphi^* g_{\mathrm{sp}}$ therefore acts by $A_H/\|\varphi\|^2$ on $\{e_+, e_-\}$ and by $A_z/\|\varphi\|^2$ on $\{e_0\}$, giving (3). Since $A_H, A_z > 0$ (established in Theorem 6.1 of [2]), the pullback is positive-definite, giving (2). □

Remark 4.6 (Status of Lemma 4.5)**.** *Items (1) and the positivity assertion of (2) are structural and hold unconditionally. Item (3) and the full form of (2) are conditional on the existence of the equivariant bridge φ (i.e. on the truth of Conjecture 4.7 below) and on Theorem 6.1 of [2] (which is itself conditional on [H-lift]). The lemma is therefore to be read as:* if the bridge exists, it is isometric in the graded sense *— and in that case the spatial metric is uniquely determined up to the two constants A_H and A_z.*

4.5 The Bridge Conjecture

The results above constrain the form of any equivariant bridge. We now state the remaining open question as a falsifiable conjecture.

Conjecture 4.7 (Bridge existence)**.** *There exists an $\mathfrak{su}(2)$-equivariant isomorphism*

$$\varphi : H_{\mathrm{eff}} = \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}, \tag{2}$$

where $W_{\mathrm{sp}} = \mathrm{span}\{\tilde{X}, \tilde{Y}, \tilde{Z}\} = \mathfrak{heis}_3/\mathbb{R}\tau$ is the spatial sector of the effective metric, compatible with the Schrödinger representation in the following sense. The compatibility is understood at the level of quadratic forms induced by the generators, not at the level of Lie algebra homomorphisms (which do not exist, by Proposition 3.1). Concretely: the quadratic form $Q_{\mathcal{C}} = 2\|\cdot\|^2$ on $\mathrm{Sym}^2(V_\rho)$, when transported to W_{sp} via φ, matches the spatial block of $\sigma_2(L_{\mathrm{eff}})$ restricted to the admissible sector of $L^2(\mathbb{R})$ in the large-q limit of Q5a [3].

By Lemma 4.3, if this map exists it is unique up to scalar. By Lemma 4.5, if it exists the spatial metric g_{sp} is positive-definite and takes the graded form (3), with A_H controlling the horizontal sector and A_z controlling the central sector. The conjecture is therefore not an additional degree of freedom: it is a binary existence question whose answer either closes the identification of the two "3"s or definitively separates them.

5 Falsifiability Criterion

The Rigidity Lemma 4.3 reduces the question to a single binary test. Since the bridge, if it exists, is unique up to scalar, there is no ambiguity about what to test.

Criterion 5.1 (Symbol restriction to $\mathrm{Sym}^2(V_\rho)$)**.** *Let $\sigma_2(L_{\mathrm{eff}})$ be the principal symbol of the effective operator L_{eff} on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$, evaluated on the spatial block (k_X, k_Y, k_Z). Via the grading correspondence of Remark 4.2, identify the spatial cotangent directions (k_X, k_Y, k_Z) with the dual basis of $\mathrm{Sym}^2(V_\rho)$ via the spin-weight assignment $k_X \leftrightarrow e_+$, $k_Y \leftrightarrow e_-$, $k_Z \leftrightarrow e_0$. Test whether:*

$$\sigma_2(L_{\mathrm{eff}})\big|_{\mathrm{Sym}^2(V_\rho)} = A_H(k_X^2 + k_Y^2) + A_z k_Z^2, \quad A_H, A_z > 0. \tag{3}$$

If this identity holds (with A_z arising from sub-principal corrections as in Q5b-O2), then Conjecture 4.7 is confirmed, the two "3"s are identified, and Q5b-O2 is resolved. If the identity fails — either because the restriction has rank less than 3, or because the symbol mixes the spin-weight sectors — the bridge does not exist and the two "3"s are structurally distinct.

Remark 5.2 (Computability and partial analytical verification)**.** *Criterion 5.1 has now been partially resolved. Section 6 proves analytically that $[F_c, L_{\mathrm{Weil}}] = 0$ (Proposition 6.1), which implies that the cross terms $k_X k_Z$ and $k_Y k_Z$ vanish structurally for all (q, c) (Corollary 6.2). Numerical verification on O25 checkpoints for $q \in \{61, 101, 151\}$ confirms this for $c \in \{3, 5, 7\}$ (Table 1). The remaining open part of Criterion 5.1 is the isotropy condition $A_H = A_z$: whether the two horizontal directions and the central direction carry equal effective-metric coefficients requires the sub-principal analysis of Q5b-O2 of [2].*

6 Numerical Verification via O25 Checkpoints

6.1 Metaplectic symmetry of the discrete Weil Laplacian

The Weil Laplacian L_{Weil} on $\mathbb{C}^q$ for character c is

$$L_{\mathrm{Weil}} = 4I - W_a - W_a^\dagger - W_b - W_b^\dagger, \tag{4}$$

where W_a is the cyclic shift $(W_a f)(k) = f(k-1 \bmod q)$ and $W_b = \mathrm{diag}(e^{2\pi i c k/q})_{k=0}^{q-1}$ is the character-c phase operator. The chirped discrete Fourier transform F_c is defined by

$$(F_c)_{jk} = \frac{1}{\sqrt{q}}\, e^{2\pi i c j k/q}, \qquad j,k \in \mathbb{Z}/q\mathbb{Z}. \tag{5}$$

It is the metaplectic representative of the symplectic rotation $(x,\xi) \mapsto (-\xi, x)$ in the character-c Weil representation: $F_c W_a F_c^\dagger = W_b^\dagger$ and $F_c W_b F_c^\dagger = W_a^\dagger$.

Proposition 6.1 (Metaplectic symmetry)**.** *For every odd prime q and every c with* $\gcd(c,q) = 1$*, the chirped DFT F_c commutes with the Weil Laplacian:*

$$[F_c,\, L_{\mathrm{Weil}}] = 0.$$

Proof. From the defining properties of F_c:

$$F_c\, L_{\mathrm{Weil}}\, F_c^\dagger = 4I - F_c W_a F_c^\dagger - F_c W_a^\dagger F_c^\dagger - F_c W_b F_c^\dagger - F_c W_b^\dagger F_c^\dagger = 4I - W_b^\dagger - W_b - W_a^\dagger - W_a = L_{\mathrm{Weil}}.$$

Hence $F_c L_{\mathrm{Weil}} = L_{\mathrm{Weil}} F_c$. □

Corollary 6.2 (Block-diagonal structure)**.** *The matrix L_{Weil} is block-diagonal in every F_c-eigenbasis. Consequently, for any admissible basis $B_{\mathrm{eff}} \in \mathbb{C}^{3\times q}$ whose rows lie in distinct F_c-eigenspaces, the restriction $\tilde{L} = B_{\mathrm{eff}}\, L_{\mathrm{Weil}}\, B_{\mathrm{eff}}^\dagger$ has no off-diagonal entries between sectors of different F_c-weight. In particular, the cross terms $k_X k_Z$ and $k_Y k_Z$ in $\sigma_2(L_{\mathrm{eff}})$ are* structurally zero*, independently of q and c.*

Corollary 6.2 proves the "no cross terms" part of Criterion 5.1 analytically. This holds for all (q,c) with $\gcd(c,q) = 1$; verification by computation is therefore in principle redundant for the cross terms, and the numerical campaign below confirms it.

6.2 Numerical validation

We implement Criterion 5.1 numerically using the Q5a-O5 checkpoints produced by the O25 pipeline [7]. For each prime $q \in \{61, 101, 151\}$ and each tested character c, the checkpoint stores: the GS admissible basis matrix $B_{\mathrm{eff}} \in \mathbb{C}^{q\times q}$ (first three rows give the three admissible directions), and the per-shell H_{eff}-projections $\pi_c(v^{(k)}) \in \mathbb{C}^3$ for k in the fitting window $[n_0, n_1]$.

The test proceeds in three stages. Stage A computes the per-pair covariance $C_c = N^{-1}\sum_j w_j w_j^\dagger$ in $H_{\mathrm{eff}} = \mathbb{C}^3$ and checks for a dominant direction. Stage B checks that the F_c-eigenvalue phases of $\tilde{F} = B_{\mathrm{eff}} F_c B_{\mathrm{eff}}^\dagger$ are multiples of 90° (consistent with $F_c^4 = I$). Stage C computes $\tilde{L} = B_{\mathrm{eff}} L_{\mathrm{Weil}} B_{\mathrm{eff}}^\dagger$ and measures the cross-term ratio $\rho_{Z|XY} = \max(|\tilde{L}_{01}|, |\tilde{L}_{02}|)/\bar{\lambda}$ and the XY-isotropy ratio $\rho_{XY} = |\tilde{L}_{11} - \tilde{L}_{22}|/|\tilde{L}_{11}|$, where $\bar{\lambda}$ is the mean diagonal entry.

The main finding is the systematic vanishing of $\rho_{Z|XY}$ across all tested pairs. This is consistent with Corollary 6.2 and confirms that the no-cross-terms prediction of Criterion 5.1 holds numerically for $q \in \{61, 101, 151\}$.

Table 1: Numerical test of Criterion 5.1 on O25 checkpoints. $A_z = \tilde{L}_{00}$ (Z-direction eigenvalue), $A_H = \frac{1}{2}(\tilde{L}_{11} + \tilde{L}_{22})$ (mean XY eigenvalue), $\rho_{Z|XY}$ and ρ_{XY} are relative cross-term ratios. Threshold: $\rho < 0.05$ for PASS. Footnotes: * finite-size artifact at $q = 61$ ($n_1 = 7$); † c=5 high-energy sector anomaly (see Remark 6.5).

q	c	A_z	A_H	$\rho_{Z\|XY}$	ρ_{XY}	Crit. 5.1
61	3	2.000	2.095	0.485	0.000	FAIL*
61	5	2.000	5.976	0.000	0.000	PASS
61	7	2.000	2.042	0.000	0.000	PASS
101	3	2.096	2.048	0.000	0.005	PASS
101	5	2.000	5.991	0.000	0.000	PASS
101	7	2.000	2.035	0.000	0.000	PASS
151	3	2.000	2.084	0.000	0.000	PASS
151	5	2.000	5.989	0.000	0.057	FAIL†
151	7	2.000	2.028	0.000	0.000	PASS

Remark 6.3 (Universal value $A_z = 2$ and the $\mathfrak{su}(2)$ Casimir)**.** *The Z-direction eigenvalue $A_z = \tilde{L}_{00}$ equals* 2.000 *to within numerical precision for all tested pairs except $(q, c) = (101, 3)$, where $A_z = 2.096$, consistent with a finite-window correction decreasing with q. The value $A_z = 2$ carries independent representation-theoretic significance: the $\mathfrak{su}(2)$ Casimir operator $\mathcal{C} = J_1^2 + J_2^2 + J_3^2$ acts on the spin-*1 *representation* $\mathrm{Sym}^2(V_\rho)$ *as* $j(j+1) \cdot \mathrm{Id} = 2 \cdot \mathrm{Id}$. *The numerical coincidence $A_z = C_{\mathfrak{su}(2)} = 2$ is a spectral signature consistent with the bridge: it suggests that the effective Laplacian $\tilde{L}$ restricted to H_{eff} acts, on the Z-direction, precisely as the Casimir of the $\mathfrak{su}(2)$-module* $\mathrm{Sym}^2(V_\rho)$.

Remark 6.4 (Finite-size anomaly at $(q, c) = (61, 3)$)**.** *The cross-term $\rho_{Z|XY} = 0.485$ at $(q, c) = (61, 3)$ is an outlier. At $q = 61$ the auto-calibrated window is $[n_0, n_1] = [2, 7]$, the shortest in the tested range. For $c = 3$ at $q = 101$ and $q = 151$ the same cross-term vanishes ($\rho_{Z|XY} < 0.001$). The failure at $q = 61$ is therefore a finite-size artifact of the short window, not a structural violation of Criterion 5.1.*

Remark 6.5 (The $c = 5$ high-energy sector is outside the bridge test)**.** *For all tested primes, the character $c = 5$ gives $A_H \approx 6 \approx 4 + 2$, markedly larger than $A_z = 2$. The value 6 is the graph-Laplacian eigenvalue of an anti-frequency mode (momentum $k \approx q/2$), indicating that the first three GS-independent Weil fingerprint vectors for $c = 5$ lie in the* high-energy sector *of L_{Weil}, not the low-energy sector. The physical spatial directions correspond to the low-energy sector (minimum L_{Weil} eigenvalues), which is selected by the BFS for $c \in \{3, 7\}$ but not for $c = 5$. The $c = 5$ admissible subspace is therefore outside the scope of the bridge test and does not constitute evidence against Conjecture 4.7.*

Table 2: Asymptotic isotropy data for the low-energy sector ($c \in \{3, 7\}$). $\Delta(q) = |A_H(q) - A_z(q)|$. For $c = 7$: OLS power-law fit over $q \in \{61, 101, 151\}$ gives $\Delta(q) \approx 0.264 \cdot q^{-0.44}$ ($R^2 \approx 1.00$, three points). Extrapolated values are shown for reference only.

q	A_z	A_H ($c = 7$)	Δ ($c = 7$)	A_H ($c = 3$)
61	2.000	2.042	0.042	2.095
101	2.000	2.035	0.035	2.048*
151	2.000	2.028	0.028	2.084
211	—	0.025 (extrap.)	—	—
307	—	0.021 (extrap.)	—	—

* At $(q, c) = (101, 3)$ the measured $A_z = 2.096$; Δ is computed as $|A_H - A_z|$ with this value.

No deviation from monotone decay of $\Delta(q) = |A_H(q) - A_z(q)|$ is observed for $c = 7$ over the tested range $q \in \{61, 101, 151\}$.

Conjecture 6.6 (Asymptotic isotropy). *For $c \in \{3, 7\}$ (low-energy sector), the isotropy gap converges:*

$$|A_H(q) - A_z(q)| \to 0 \quad \text{as } q \to \infty.$$

The empirical power-law fit for $c = 7$ over $q \in \{61, 101, 151\}$ gives $|A_H - A_z| \approx 0.264 \cdot q^{-0.44}$, consistent with $q^{-1/2}$ convergence within the available data range. Verification of the conjecture at $q \in \{211, 307, 401\}$ would confirm the trend and distinguish the exponent from $-1/2$.

Corollary 6.7 (Status of the bridge identification). *The following hold unconditionally or with strong numerical support:*

1. (Proved) *The effective symbol restricted to H_{eff} has no cross terms:*

$$\sigma_2(L_{\text{eff}})\big|_{H_{\text{eff}}} = A_H(k_X^2 + k_Y^2) + A_z k_Z^2.$$

2. (Numerical, $q \in \{61, 101, 151\}$) *The Z-sector coefficient is universal: $A_z = 2 = C_{\mathfrak{su}(2)}$.*

3. (Numerical, $q \in \{61, 101, 151\}$) *The horizontal gap satisfies $|A_H - A_z| < 0.05$ for $c \in \{3, 7\}$, decreasing with q.*

If Conjecture 6.6 holds, then $A_H = A_z = C_{\mathfrak{su}(2)} = 2$ in the large-q limit, Conjecture 4.7 is confirmed, and the three spatial dimensions are identified with the spin-1 representation of $\mathfrak{su}(2)$ with the effective spatial metric governed entirely by the Casimir:

$$g_{\text{sp}} = C_{\mathfrak{su}(2)} \cdot (dk_X^2 + dk_Y^2 + dk_Z^2).$$

Remark 6.8 (Three-case structure of the isotropy question). *Criterion 5.1 in its original form requires both the vanishing of cross terms and the equality $A_H = A_z$. The first condition is now settled (Corollary 6.2). For the second, three genuinely distinct outcomes are possible, with separate physical implications.*

1. *$A_H = A_z > 0$: the bridge exists and the effective spatial geometry is isotropic. The three spatial directions are metrically equivalent. This is the strongest conclusion: the two "3"s are fully identified.*

2. *$A_H \neq A_z$, both positive: the bridge exists but the effective spatial geometry is anisotropic. The three spatial directions are metrically distinct (A_H for the horizontal, A_z for the central). This is still a positive result: the diagonal structure is confirmed and the bridge is real, but the spatial metric is genuinely non-isotropic.*

3. *$A_z \leq 0$ or structurally unstable: the bridge does not exist. The two "3"s are structurally independent.*

The numerical data (Table 1) support case (1) or case (2) for $c \in \{3, 7\}$: $A_z \approx 2$ and $A_H \approx 2$ with $|A_H - A_z|/A_z < 0.05$ for all tested (q, c) in the low-energy sector, pointing strongly to case (1). Case (3) is not observed.

7 Discussion

7.1 What This Paper Establishes

The following results are proved unconditionally or structurally in this paper:

- The two occurrences of 3 in the programme arise via distinct mechanisms at distinct algebraic levels and cannot be identified by a Lie algebra isomorphism (Proposition 3.1).
- The admissible projection space $H_{\text{eff}} = \mathbb{C}^3$ carries the structure of the unique irreducible three-dimensional $\mathfrak{su}(2)$-module $\mathrm{Sym}^2(V_\rho)$, inferred from O29 (Proposition 4.1).
- Any equivariant bridge between $\mathrm{Sym}^2(V_\rho)$ and the spatial sector is unique up to positive scalar (Lemma 4.3): the bridge is rigid, not a matter of convention.
- The quadratic form induced by the $\mathfrak{su}(2)$-Casimir on $\mathrm{Sym}^2(V_\rho)$ pulls back, under any equivariant bridge, to a positive-definite graded form on the spatial sector that matches g_{sp} up to the two constants A_H and A_z (Lemma 4.5).
- The grading correspondence (Remark 4.2) connects the Carnot degree-$(1, 2)$ decomposition of $\mathfrak{heis}_3$ to the spin-weight-$(\pm 1, 0)$ decomposition of $\mathrm{Sym}^2(V_\rho)$, explaining why the $\tilde{Z}$ direction enters at sub-principal order in the effective metric.
- The no-cross-terms part of this criterion is proved analytically (Proposition 6.1 and Corollary 6.2) and confirmed numerically on O25 checkpoints (Table 1). The remaining open part is the isotropy condition $A_H = A_z$, with numerical evidence pointing to case (1) of Remark 6.8: $A_H \approx A_z \approx 2$ in the low-energy sector, decreasing gap (Table 2), and a power-law fit consistent with convergence $A_H \to A_z$ as $q \to \infty$ (Conjecture 6.6).
- The universal value $A_z = 2 = C_{\mathfrak{su}(2)}$ is a spectral signature of the bridge: it matches the eigenvalue of the $\mathfrak{su}(2)$ Casimir on $\mathrm{Sym}^2(V_\rho)$ (Remark 6.3). If Conjecture 6.6 holds, the bridge identification reduces to the statement that the effective spatial metric is governed by the $\mathfrak{su}(2)$ Casimir (Corollary 6.7).

7.2 Relation to Open Problem Q5b-O2

Open problem Q5b-O2 of [2] asks for the extension of the effective metric to a full 4×4 Lorentzian tensor including the $\tilde{Z}$-direction, via sub-principal corrections to the symbol of L_{eff}. The present paper provides the representation-theoretic prediction: if the bridge exists, the $\tilde{Z}$-direction contributes a term $A_z k_Z^2$ to the spatial block of the symbol, without cross terms with (k_X, k_Y). The absence of cross terms is no longer a prediction but a theorem (Proposition 6.1). The target for Q5b-O2 is therefore sharpened: compute the sub-principal symbol in the $\tilde{Z}$ direction and determine whether A_z equals the Casimir value $C_{\mathfrak{su}(2)} = 2$ (Remark 6.3). Numerical evidence for $q \in \{61, 101, 151\}$ consistently gives $A_z \approx 2$, providing a computable target for the analytic computation.

7.3 If the Identification Succeeds

If Criterion 5.1 holds, the programme would have derived a structural identification candidate between the three spatial directions of the effective geometry and the three-dimensional irreducible representation of the minimal admissible non-commutative algebra $\mathbb{H}$. The three-dimensionality of space would follow from the uniqueness of the spin-1 representation of $\mathfrak{su}(2)$ (itself forced by Born–Infeld admissibility, O23 [8]). The tag `spt` would then be retroactively justified on O23, O28, and O29.

7.4 If the Identification Fails

A failure of Criterion 5.1 — either through rank deficiency or through cross terms mixing the spin-weight sectors — would imply that H_{eff} is an internal representation space not identifiable with the spatial sector of the effective geometry. This would be a significant negative result: it would bound the scope of the O-series, establish that the two “3”s are structurally independent coincidences, and sharpen the open problem for a future paper.

8 Conclusion

The programme exhibits two independent structural outputs equal to the integer 3: the Carnot spatial sector of $\mathrm{Heis}_3(\mathbb{R})$ via the Bass–Guivarch homogeneous dimension, and the admissible effective space $H_{\mathrm{eff}} = \mathbb{C}^3$ via quaternionic minimality and the symmetric rank formula. These two “3”s are numerically equal but algebraically distinct: $\mathfrak{su}(2) \not\simeq \mathfrak{heis}_3$. The present paper establishes three structural results that constrain any possible bridge between them. First, $H_{\mathrm{eff}} \simeq \mathrm{Sym}^2(V_\rho)$ as an $\mathfrak{su}(2)$-module, making the admissible “3” the dimension of the spin-1 representation. Second, the irreducibility of $\mathrm{Sym}^2(V_\rho)$ forces any equivariant bridge to be unique up to scalar (Rigidity Lemma), removing all conventional freedom from the identification. Third, the $\mathfrak{su}(2)$-Casimir pulls back to a positive-definite graded quadratic form on the spatial sector, matching the structure of the effective metric g_{sp} with the spin-weight grading reproducing the Carnot grading of $\mathfrak{heis}_3$ (Schrödinger Quadratic Form Lemma). The open question is thereby reduced to a single computable criterion (Criterion 5.1) on the principal symbol of L_{eff}, which simultaneously constitutes a resolution strategy for open problem Q5b-O2 of [2]. The no-cross-terms part of this criterion is now settled: we prove analytically that $[F_c, L_{\mathrm{Weil}}] = 0$ for all (q, c) (Proposition 6.1), which forces the cross terms between spin-weight sectors to vanish structurally; numerical verification on O25 checkpoints for $q \in \{61, 101, 151\}$ is consistent with this result (Table 1). The remaining open question is whether $A_H = A_z$ (isotropy condition, Conjecture 6.6). Numerical evidence gives $A_z \approx A_H \approx 2$ in the low-energy sector ($c \in \{3, 7\}$), with the universal value $A_z = 2$ matching the $\mathfrak{su}(2)$ Casimir eigenvalue on $\mathrm{Sym}^2(V_\rho)$ (Remark 6.3). A power-law fit of $|A_H - A_z|$ over $q \in \{61, 101, 151\}$ for $c = 7$ gives an exponent ≈ -0.44, consistent with convergence to zero as $q \to \infty$ (Table 2). The identification of the two “3”s is thereby reduced to an asymptotic scalar equality for which all current analytical and numerical evidence is positive; verification at $q \in \{211, 307, 401\}$ would establish that the effective spatial metric is governed entirely by the Casimir $C_{\mathfrak{su}(2)} = 2$ (Corollary 6.7).

References

[1] J. Beau. Asymptotic calibration of the bfs window and effective dimension of the admissible trajectory, 2026. O28 paper.

[2] J. Beau. Bfs shell stratification and the emergence of four-dimensional lorentzian geometry, 2026. spectral admissibility subprogram "Q5b" paper.

[3] J. Beau. Large-q limit of the admissible fibre: Representational and operator convergence, 2026. spectral admissibility subprogram "Q5a" paper.

[4] J. Beau. Quaternionic rigidity of admissible morphisms: Every admissible $\phi_{q,\rho}$ necessarily factors through $\mathfrak{su}(2)$, 2026. O27 paper.

[5] J. Beau. Spectral capacity functional and the binary-polyhedral maximality conjecture, 2026. Step B of Cosmochrony Spectral Admissibility sub-Program.

[6] J. Beau. Spin-$\frac{1}{2}$ sector identification via the symmetric rank formula: Effective dimension of the admissible covariance in $\mathrm{End}(v_\rho)$, 2026. O29 paper.

[7] J. Beau. Systematic pair-level campaign for δ_{pair}: Convergence, inter-pair concentration, and normalization structure, 2026. O25 paper.

[8] J. Beau. Three stable directions from quaternionic minimality: Derivation of $\sigma_c(n_3) = 3$ from born–infeld fibre admissibility, 2026. Spectral admissibility sub-program "O23" paper.